Hardening objective

Convert the paper from:

- a strong diagnostic note

into:

- a theorem-guided flagship manuscript whose central dependence claim is visibly forced.

The main vulnerabilities to close are:

1. the two-case exhaustion is stated but not visibly proven,
2. Everettian dependence on reference fixation is asserted more than derived,
3. decision-theoretic and self-locating replies are discussed but not structurally neutralized,
4. the semantic infrastructure constraint risks looking ad hoc,
5. confirmation collapse is announced without a compact formal model, and
6. layer separation is not strict enough.

v1 → v2 patch philosophy

Do **not** rewrite the paper into later-corpus language.
Do **not** turn the main text into an admissibility treatise.
Do **not** bloat the prose with literature until the theorem spine is closed.

Instead:

- keep the current title,
- preserve the original insight,
- insert a visible theorem ladder,
- strengthen local transitions,
- add one formal toy model,
- and harden each vulnerable paragraph with exact role.

Patch map by section

Patch 1 — Abstract: convert diagnosis into theorem-bearing claim

v1 problem

The abstract states the central move clearly, but it still reads as a high-level diagnosis rather than a manuscript with a visible forcing result.

v2 patch

Replace the abstract with a four-move structure:

1. standard measurement-problem framing,
2. exact retyping claim,
3. theorem-bearing mechanism,
4. interpretation-level consequence.

exact content to insert

Open with something like:

> This paper argues that the measurement problem is structurally prior to outcome selection: if a physically non-selective theory is used to underwrite truth-apt diachronic measurement discourse, then it must supply a principle of reference stabilization for measurement records.

Then add:

> The paper proves a two-case exhaustion for admissible evidential reference assignments: either admissible assignments preserve the same evidential proposition up to confirmational-role equivalence, or they permit divergent confirmational roles, in which case theory-level confirmation is not well-defined without additional restriction.

Then close with the interpretation taxonomy sentence:

> Collapse and hidden-variable theories secure singular reference physically; Everettian approaches require non-dynamical restriction of admissible reference relations; instrumentalist strategies avoid the problem by declining the target discourse.

reason

This single patch reclassifies the paper from “interesting reframing” to “exact conditional theorem claim.”

Patch 2 — Introduction opening: sharpen the target conditional

v1 problem

The introduction says the central claim is conditional, but the conditional is not yet promoted to the paper’s governing premise.

v2 patch

Insert a boxed thesis paragraph at the end of the opening section.

exact theorem-style thesis box

Add after the paragraph ending with “divergent strategies adopted by competing interpretations”:

Thesis.
Let TTT be a physically non-selective theory used to underwrite truth-apt diachronic discourse about measurement, memory, anticipation, and evidential confirmation. Then TTT must either:

1. fix admissible reference assignments for measurement records sufficiently to preserve confirmational role, or
2. fail to underwrite theory-level confirmation and diachronic first-person discourse in the sense presupposed by explanatory practice.

reason

This gives the manuscript a governing formal sentence early, exactly as Phase 1 required.

Patch 3 — Introduction: separate the three levels explicitly

v1 problem

The paper still risks allowing a referee to accuse it of conflating physical, semantic, and epistemic questions.

v2 patch

Insert a short three-level clarification near the end of the introduction.

exact content

Add a paragraph:

> Three levels must be distinguished throughout. First, the **physical level**, which concerns the ontology and dynamics delivered by the theory. Second, the **referential level**, which concerns what branch-history or record is the referent of diachronic discourse. Third, the **confirmational level**, which concerns how such records function as evidence for or against rival hypotheses. The paper's central claim is that physically non-selective theories may settle the first level while leaving the second underdetermined, and that instability at the second level propagates to the third.

reason

This blocks the "semantics/physics conflation" objection before it starts.

Patch 4 — Introduction: harden the two-case exhaustion preview

v1 problem

The "two-case exhaustion" is excellent but too compressed and insufficiently formal in the present note.

v2 patch

Retain the heading but reframe it as a theorem preview.

exact replacement

Replace the current mini-heading paragraph with:

Theorem-preview (Two-case exhaustion for evidential reference).
Suppose a theory is used to underwrite theory-level claims about what was observed and how those observations confirm or disconfirm rival hypotheses. Then admissible reference assignments for measurement records fall into exactly one of two classes:

1. **role-preserving class**: all admissible assignments determine the same evidential proposition, or differ only by a role-preserving equivalence that leaves confirmational function unchanged;
2. **role-divergent class**: admissible assignments differ in confirmational role, including by changing comparative support or reversing hypothesis ordering.

No third class exists. In the second case, theory-level confirmation is reference-relative rather than theory-relative and so is not well-defined without further restriction on admissible reference.

reason

This creates the exact sentence the later theorem must prove.

Patch 5 — Section 2: define “physically non-selective theory” formally

v1 problem

The phrase is intuitively clear, but not yet definitionally locked.

v2 patch

Insert a definition at the beginning of Section 2.

exact definition

Definition 2.1 (Physical non-selection).
A theory is physically non-selective with respect to a measurement process if, after a measurement interaction, the theory's physical description realizes multiple successor histories containing stable measurement records and physically continuous observer-successors, while supplying no physical principle that uniquely privileges one such history as the diachronic continuation relevant to pre-measurement first-person discourse.

reason

This turns the Everett target from a vibe into a typed object.

Patch 6 — Section 2: add minimal branching model

v1 problem

The Everett discussion remains too prose-heavy.

v2 patch

Add a compact model with notation.

exact structure

Insert:

- pre-measurement observer O0O_0O0,
- system SSS in superposition ∑ici∣si⟩\sum_i c_i \lvert s_i \rangle∑ici∣si⟩,
- apparatus AAA,
- post-measurement state
  ∑ici∣Oi⟩∣Ai⟩∣si⟩,\sum_i c_i \lvert O_i \rangle \lvert A_i \rangle \lvert s_i \rangle,i∑ci∣Oi⟩∣Ai⟩∣si⟩,
- with decoherence suppressing interference among macroscopically stable record states.

Then state:

Lemma 2.2 (Branch-local determinacy).
Each decohered branch contains a determinate record and a determinate branch-relative observer-state.

Lemma 2.3 (Non-selection).
The physical structure above does not by itself determine which OiO_iOi is the referent of pre-measurement diachronic first-person discourse.

proof sketch to include

For Lemma 2.3:

- the theory gives multiple physically realized successors,
- each inherits physical continuity up to branching,
- no extra physical selector is present,
- therefore branch multiplicity plus continuity does not yield singular diachronic referent.

reason

This is the first place where the Everett critique becomes unavoidable rather than rhetorical.

Patch 7 — Section 3: define admissible reference assignment and role-preserving equivalence

v1 problem

The semantic core is still schematic.

v2 patch

Add three definitions.

exact definitions

Definition 3.1 (Measurement record).
A measurement record is a physically realized structure used within the theory's explanatory practice to anchor claims about what outcome was observed.

Definition 3.2 (Admissible reference assignment).
An admissible reference assignment is a rule mapping diachronic measurement discourse to candidate record-histories in a way compatible with the theory's ontology and the discourse class under consideration.

Definition 3.3 (Confirmational-role equivalence).
Two admissible reference assignments are confirmationally equivalent if every theory-level evidential claim licensed under one assignment has the same comparative confirmational role under the other.

reason

This gives the theorem in Section 4 a clean vocabulary.

Patch 8 — Section 4: make the exhaustion theorem explicit and prove it

v1 problem

This is the paper's hinge and currently its biggest referee vulnerability.

v2 patch

Turn the section into a theorem section with a visible proof.

exact theorem

Theorem 4.1 (Two-case exhaustion for admissible evidential reference).
Let RRR be the class of admissible reference assignments for a theory's measurement records. Then exactly one of the following holds:

1. all assignments in RRR are pairwise confirmationally equivalent;
2. there exist assignments in RRR that are not confirmationally equivalent.

No third case exists.

proof

Trivial set-theoretic exhaustion:

- either the equivalence relation is universal on RRR,
- or it is not.
  But then add the crucial interpretive sentence:
- the nontrivial work is not the logical disjunction,
- it is the consequence theorem below.

exact corollary

Corollary 4.2 (Confirmation-collapse corollary).
If case (2) holds and admissible assignments alter comparative evidential role, then theory-level confirmation is reference-relative rather than theory-relative.

reason

The exhaustion theorem itself is simple. The real work is making the consequence theorem non-optional.

Patch 9 — Section 4: add a compact toy model of hypothesis-order reversal

v1 problem

You announce confirmational collapse, but a hostile referee can still say “show me.”

v2 patch

Add a toy model with two hypotheses and two admissible assignments.

exact model shape

Let H1,H2H_1, H_2H1,H2 be rival hypotheses.
Let mmm be a measurement event.
Let admissible assignments r1,r2r_1, r_2r1,r2 map the same pre-measurement discourse to different realized records M1,M2M_1, M_2M1,M2.
Suppose:

- under r1r_1r1, M1M_1M1 favors H1H_1H1 over H2H_2H2,
- under r2r_2r2, M2M_2M2 favors H2H_2H2 over H1H_1H1.

Then:

Proposition 4.3 (Hypothesis-order instability).
If both r1r_1r1 and r2r_2r2 are admissible and not confirmationally equivalent, comparative confirmation is not fixed by the theory alone.

reason

This single worked example will do more than several pages of abstract warning.

Patch 10 — Section 5: explicitly separate branch-relative definiteness from diachronic referential determinacy

v1 problem

This section is already strong, but it must do more work because this is the first obvious Everett reply.

v2 patch

Upgrade the clarificatory note into a theoremized distinction section.

exact lemma

Lemma 5.1 (Branch-relative determinacy does not entail diachronic determinacy).
The existence of determinate branch-relative post-measurement experiences does not determine which such branch-relative experience is the referent of pre-measurement diachronic first-person discourse.

proof idea

- branch-local determinacy concerns within-branch content,
- diachronic determinacy concerns cross-branch successor-reference,
- the former can hold while the latter remains underdetermined.

reason

This closes the “but each branch observer is definite” objection formally.

Patch 11 — Section 5 or 7: neutralize decision theory structurally

v1 problem

The present discussion is correct, but a hostile referee will demand more than prose.

v2 patch

Add one theorem.

exact theorem

Theorem 5.2 (Decision-theoretic dependence).
Any decision-theoretic framework that treats pre-measurement anticipation, preference updating, or branch-indexed utility as truth-apt and action-guiding presupposes a determinate admissible relation between the pre-measurement subject and future record-bearing successors.

proof strategy

- decision requires that preference be about determinate future outcomes,
- branch-indexed utility assignments are not action-guiding unless the subject’s future-successor relation is sufficiently fixed,
- therefore decision theory does not replace reference fixation; it presupposes it.

reason

This is one of the most important hostile-referee blocks.

Patch 12 — Section 6: derive the semantic infrastructure constraint instead of merely introducing it

v1 problem

Right now this section risks looking like a semantic plug-in.

v2 patch

Open the section with a necessity theorem.

exact theorem

Theorem 6.1 (Reference-stabilization necessity).
If a physically non-selective theory is used to underwrite truth-apt diachronic discourse whose records must function as evidential anchors, then the theory or its admissible interpretive completion must restrict admissible reference assignments sufficiently to preserve confirmational role.

proof

From Theorem 4.1 and Corollary 4.2:

- if no restriction exists, role-divergent assignments remain admissible,
- then confirmation is reference-relative,
- but explanatory practice presupposes theory-level confirmation,
- therefore some reference-stabilizing restriction is necessary.

reason

This turns “semantic infrastructure” from an optional proposal into a forced consequence.

Patch 13 — Section 6: add explicit anti-ad-hoc paragraph

v1 problem

The “ad hoc semantics” objection remains alive.

v2 patch

Insert a paragraph:

> The semantic infrastructure constraint is not introduced as an extra metaphysical postulate whose only purpose is to save a preferred interpretation. It is derived from the conditional use of the theory itself. If the theory is asked to underwrite truth-apt diachronic discourse and theory-level confirmation, then admissible reference must be restricted enough to preserve those roles. The constraint is therefore forced by explanatory use, not appended from outside it.

reason

This should be in the main text, not buried in replies.

Patch 14 — Sections 7–10: convert interpretive survey into selector taxonomy

v1 problem

These sections are good but still a bit essayistic.

v2 patch

Introduce one formal classification theorem before the survey.

exact theorem

Theorem 8.1 (Selector taxonomy).
Given the standing conditional of the paper, any interpretation of quantum measurement must fall into one of four classes:

1. **dynamic selector**: singular reference fixed by dynamics,
2. **ontic selector**: singular reference fixed by added ontology,

3. **semantic selector**: singular reference fixed by non-dynamical restriction of admissible reference assignments,
4. **discourse deflation**: the theory is not taken to underwrite the target diachronic discourse.

Then classify:

- collapse = class 1,
- hidden variable = class 2,
- Everett + semantic infrastructure = class 3,
- QBism/instrumentalism = class 4.

reason

This preserves neutrality and gives the paper a memorable output.

Patch 15 — New section before conclusion: explicit non-claims

v1 problem

You state non-claims, but they need to be gathered and frozen.

v2 patch

Add a section titled:

What the paper does not claim

List explicitly:

- no derivation of the Born rule,
- no modification of quantum dynamics,
- no refutation of Everettian physics,
- no proof that all interpretations are false except one,
- no corpus dependence in the main theorem ladder.

reason

This removes a huge amount of referee slippage.

Patch 16 — New appendix: theorem dependency map

v1 problem

Submission-grade flagship paper needs auditability.

v2 patch

Appendix A:

- dependency graph:
  - Def. 2.1 → Lemma 2.3
  - Defs. 3.1–3.3 → Theorem 4.1
  - Theorem 4.1 → Cor. 4.2
  - Lemma 5.1 + Cor. 4.2 → Theorem 6.1
  - Theorem 6.1 + Section 8 → Theorem 8.1
- one-page visual map

reason

This is cheap and very high value.

Patch 17 — New appendix: hostile-referee attack matrix

v1 problem

Phase 7 calls for this; it should now be concrete.

v2 patch

Appendix B table:

Objection	Blocked by
Branch-relative determinacy is enough	Lemma 5.1
Decision theory solves it	Theorem 5.2
Semantic constraint is ad hoc	Theorem 6.1
Everett has stable records already	Lemma 2.3 + Theorem 6.1
Reference can vary without breaking science	Corollary 4.2 + Prop. 4.3
This is just self-location	Theorem 5.2 + Section 5 distinction
Instrumentalism dissolves the problem	taxonomy class 4; explicit concession

reason

Makes rejection costlier.

Patch 18 — Conclusion: sharpen to exact accomplishment

v1 problem

The conclusion is good, but still slightly broad.

v2 patch

Replace with a structure like:

1. what was shown,
2. what follows for Everett,
3. what follows for interpretation generally,
4. what remains open.

exact closing sentence

> The paper does not show that quantum mechanics needs new dynamics. It shows that any physically non-selective theory used to underwrite truth-apt diachronic measurement discourse must secure reference stabilization somewhere, and that the measurement problem is therefore best understood as the problem of where that work is done.

reason

This is the cleanest flagship ending.

Order of execution

Run the patches in this order:

1. Patch 1–4: abstract + introduction
2. Patch 5–8: definitions + exhaustion theorem
3. Patch 9–13: confirmation model + Everett dependence + semantic necessity
4. Patch 14–15: taxonomy + non-claims
5. Patch 16–18: appendices + conclusion

Minimal v2 theorem ladder

For clarity, v2 should visibly contain at least these labeled results:

1. **Definition 2.1** — Physical non-selection
2. **Lemma 2.2** — Branch-local determinacy
3. **Lemma 2.3** — Everettian non-selection
4. **Definitions 3.1–3.3** — Record, admissible assignment, confirmational-role equivalence
5. **Theorem 4.1** — Two-case exhaustion for admissible evidential reference
6. **Corollary 4.2** — Confirmation-collapse corollary
7. **Proposition 4.3** — Hypothesis-order instability model

8. **Lemma 5.1** — Branch-relative determinacy does not entail diachronic determinacy
9. **Theorem 5.2** — Decision-theoretic dependence
10. **Theorem 6.1** — Reference-stabilization necessity
11. **Theorem 8.1** — Selector taxonomy

Bottom line

The v1 paper already contains the decisive insight:

> physically non-selective theories leave measurement-reference underdetermined.

The v2 hardening task is not to invent a new paper.
It is to force what the current note already sees.

The exact transition is:

- from diagnostic prose
- to visible dependence theorems
- to a confirmation-centered impossibility structure
- to a clean interpretation taxonomy.